

Multifractal Large Deviations in Quantum Entanglement: Ideal Fermi-Gas Mapping, Markov Tensor Cascades, and Quartic Casimir Symmetry

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Abstract

The thermodynamic formalism of multifractals assigns to any normalised distribution a mass exponent $\tau(q)$ and a singularity spectrum $f(\alpha)$. Applied to the eigenvalues of a reduced density matrix, this formalism turns Rényi entropies into generalised dimensions and the refined Rényi entropy into $f(\alpha_q)$. We use it to compare two notions that are often conflated: the multifractality of single-particle wave functions at localisation transitions, and the multifractal structure of many-body entanglement spectra.

For Gaussian fermionic states we prove that the refined Rényi entropy is exactly the entropy of an ideal Fermi gas at temperature $1/q$ whose single-particle levels are the entanglement energies. The entanglement singularity spectrum is therefore always non-negative and concave. Its dependence on wave-function multifractality is mediated entirely by the density of entanglement energies, which is controlled by inter-eigenstate overlaps rather than by single-state moments.

For tensor-network cascades with correlations between layers, modelled by a Markov chain, the mass exponent is set by a Perron root. Its first-order correction is linear in the subleading eigenvalue λ_2 , and it is real-analytic for $\lambda_2 < 1$. A first-order (Maxwell) transition appears only in the reducible limit $\lambda_2 = 1$.

At the quantum Hall transition, Weyl-group invariance makes the anomalous exponents functions of $u = q(1 - q)$. The measured non-parabolicity b_1 is then the coefficient of the u^2 (higher-Casimir) term. This term appears at four loops in the $2 + \varepsilon$ expansion with the same sign as observed, and its non-vanishing excludes theories in which the relevant operators are current-algebra primaries of a WZW model.

Finally, we compute Gauss–Bonnet cosmic branes in $d = 4$. We confirm that the second Taylor coefficient of $\tau(q)$ tracks C_T/a , find that the refined Rényi entropy turns negative at finite q for positive Gauss–Bonnet coupling, and explain why no correspondence between holographic couplings and the quantum Hall coefficient b_1 should be expected.

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1 Introduction

The multifractal formalism [1, 2] describes a normalised measure $\{\mu_i\}$ on cells of size ε through the scaling of its moments, $\sum_i \mu_i^q \sim \varepsilon^{\tau(q)}$, and through the Legendre transform $f(\alpha) = q\alpha - \tau(q)$, $\alpha = \tau'(q)$, which counts the cells with $\mu_i \approx \varepsilon^\alpha$. The formalism is thermodynamic: q plays the role of an inverse temperature, τ of a free energy, α of an energy and f of an entropy.

Two quite different physical objects are called “multifractal” in quantum many-body physics.

1. *Wave-function multifractality.* At Anderson transitions and at the integer quantum Hall (IQH) plateau transition, the single-particle probability $|\psi(r)|^2$ has moments $\langle |\psi|^{2q} \rangle / \langle |\psi|^2 \rangle^q \sim L^{-\Delta_q}$ with non-linear anomalous exponents Δ_q [3]. The relevant measure lives on real space and is averaged over disorder.
2. *Entanglement-spectrum multifractality.* The eigenvalues $\{\lambda_i\}$ of a reduced density matrix ρ_A form a normalised measure on the spectrum. Its moments are the Rényi entropies, $S_q = \frac{1}{1-q} \ln \text{Tr} \rho_A^q$. For a 1+1-dimensional CFT the full eigenvalue density is the Calabrese–Lefevre distribution [4].

The same formalism applies to both, but the measures are different, and so are their symmetries and positivity properties. Jia, Subramaniam, Gruzberg and Chakravarty showed that the summed single-site entanglement of a single-particle critical state scales as $\alpha_1 \ln L$ [5], a link through one point of the spectrum only. Dong noted that the refined Rényi entropy is a thermodynamic entropy with respect to the modular Hamiltonian [6], which places the entanglement problem squarely within the thermodynamic formalism.

Results. Building on these works, we obtain the following.

1. *Ideal Fermi-gas theorem* (Theorem 1). For any Gaussian fermionic state, the refined Rényi entropy equals the entropy of an ideal Fermi gas at temperature $1/q$ with the entanglement energies as single-particle levels. Consequently $f_{\text{ent}} \geq 0$ is concave and is fixed by the density of entanglement energies. Wave-function multifractality enters only through that density, which depends on overlaps between different eigenstates (Proposition 2.2).
2. *Markov tensor cascades* (Section 3). For layer-to-layer correlations modelled by a Markov chain, $\tau(q)$ is a Perron root. Its correction is linear in λ_2 , not a power λ_2^q , and it is analytic for $\lambda_2 < 1$. Phase transitions require reducibility or quenched averaging.

3. *Casimir structure at the IQH transition* (Section 4). Writing the measured spectrum in the Weyl-invariant variable $u = q(1 - q)$ identifies b_1 with the coefficient of u^2 . This term arises at four loops in the $2 + \varepsilon$ expansion, with the observed sign, and is incompatible with current-algebra primaries of a WZW model.
4. *Gauss–Bonnet cosmic branes* (Section 5). In a companion paper [25] we applied the cosmic-brane description to fractal entangling surfaces. There, in Einstein gravity, the shape of $f(\alpha)$ is fixed by the bulk through the horizon radius x_q , and in $d = 2$ it reduces to the Calabrese–Lefevre form, which is universal for all CFTs. Here we extend the analysis to Gauss–Bonnet gravity. The q -expansion of the holographic mass exponent is controlled by CFT data such as C_T/a . Positive Gauss–Bonnet coupling leads to a negative refined Rényi entropy at finite q . No dictionary between holographic couplings and b_1 is warranted.

Section 6 discusses limitations, and the Appendix collects numerical methods and large-deviation derivations.

2 Entanglement thermodynamics and the ideal Fermi-gas theorem

2.1 The thermodynamic dictionary

Let $\mathcal{L} \rightarrow \infty$ be a large parameter, for instance $\mathcal{L} = \ln(1/\varepsilon)$ or $\mathcal{L} = \ln L$, and suppose

$$\ln \text{Tr} \rho_A^q = -\tau(q) \mathcal{L} + o(\mathcal{L}) \quad (1)$$

on an open interval I of q on which τ is differentiable.

Proposition 2.1 (Thermodynamic dictionary). *Under (1), for $q \in I$,*

$$S_q = D_q \mathcal{L} + o(\mathcal{L}), \quad D_q = \frac{\tau(q)}{q-1}, \quad (2)$$

$$\tilde{S}_q \equiv q^2 \partial_q \left(\frac{q-1}{q} S_q \right) = f(\alpha_q) \mathcal{L} + o(\mathcal{L}), \quad f(\alpha_q) = q\alpha_q - \tau(q), \quad \alpha_q = \tau'(q), \quad (3)$$

and $\tilde{S}_q$ is the von Neumann entropy of the escort state $\rho_A^q / \text{Tr} \rho_A^q$.

Proof. $\Phi(q) = \ln \text{Tr} \rho_A^q$ is convex, being a log-sum-exp of functions linear in q . Pointwise convergence of the convex functions $\Phi/\mathcal{L}$ to a limit that is differentiable on I implies convergence of their derivatives [7, Thm. 25.7]. Since $\frac{q-1}{q} S_q = -\Phi/q$, we have $\tilde{S}_q = \Phi - q\Phi'$, which converges to $(q\tau' - \tau)\mathcal{L}$. For the escort state $\pi_i = \lambda_i^q / \sum_j \lambda_j^q$, one finds $-\sum_i \pi_i \ln \pi_i = q\langle -\ln \lambda \rangle_\pi + \Phi = \Phi - q\Phi'$. $\square$

Particular values have direct information-theoretic meaning: D_0 is the Hartley entropy (log-rank), D_1 the von Neumann entropy, D_2 the collision entropy, and $D_\infty = \alpha_{\min}$ the min-entropy, all per unit $\mathcal{L}$. Figure 1 illustrates the dictionary for a two-scale Cantor measure read as an entanglement spectrum. There ρ_A is either the exact cascade spectrum at finite resolution, or a tensor product of n “bond” qubits with Schmidt weights (p_1, p_2) . In the second case $S_q = n H_q(p)$ exactly, so $D_q = H_q(p)/\ln 2$.

2.2 The ideal Fermi-gas theorem

For a Gaussian fermionic state, Peschel’s theorem [9] gives $\rho_A = Z^{-1} \exp(-\sum_k \varepsilon_k c_k^\dagger c_k)$. The entanglement energies ε_k are related to the eigenvalues ζ_k of the correlation matrix $C_{ij} = \langle c_i^\dagger c_j \rangle$, $i, j \in A$, by $\zeta_k = (1 + e^{\varepsilon_k})^{-1}$.

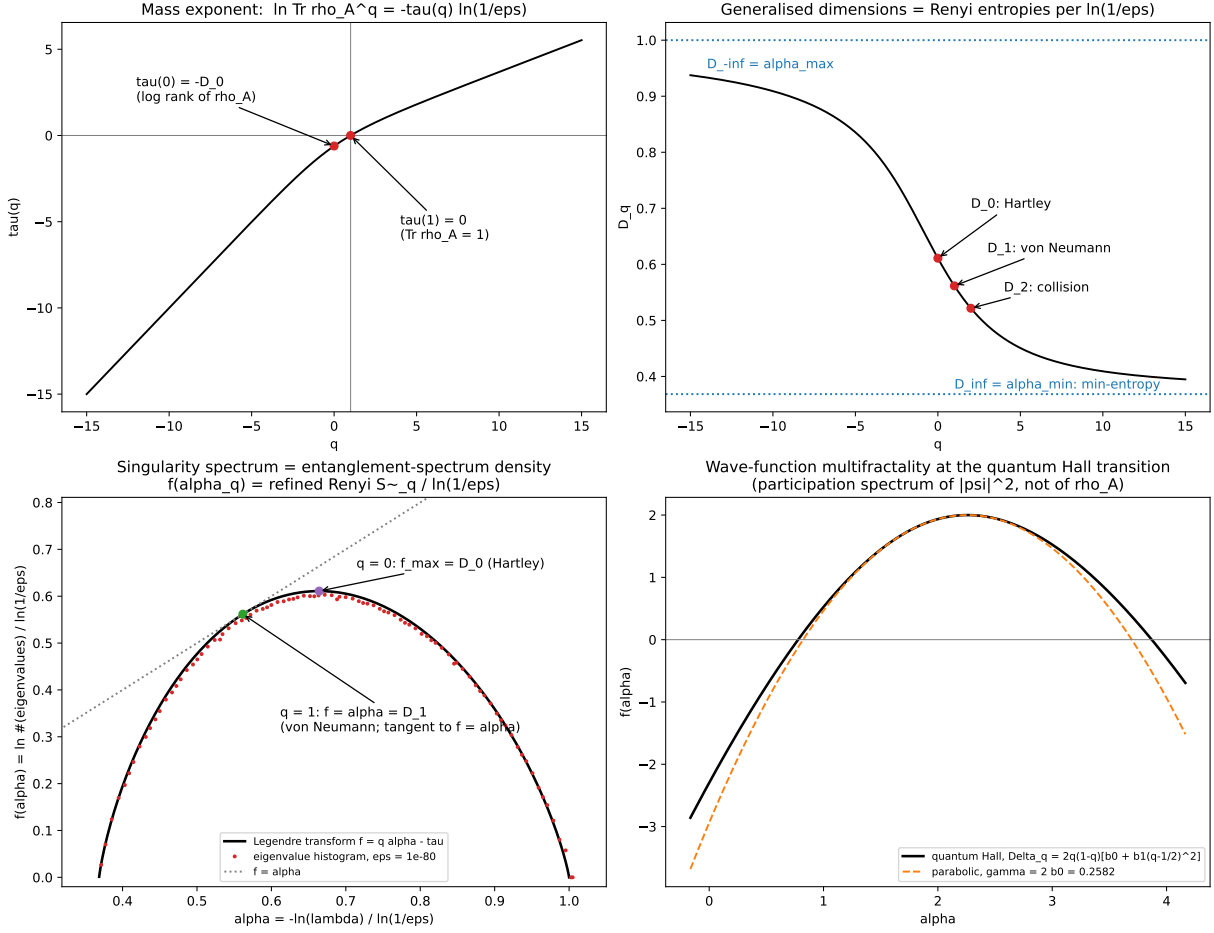


Figure 1: Thermodynamic dictionary for a two-scale Cantor measure ($p = (0.6, 0.4)$, $r = (0.25, 0.4)$) read as the eigenvalue distribution of ρ_A . Top left: mass exponent, with $\tau(0) = -D_0$ (log-rank) and $\tau(1) = 0$ (normalisation). Top right: generalised dimensions $D_q = S_q/\mathcal{L}$, with the Hartley, von Neumann, collision and min-entropies marked. Bottom left: singularity spectrum from the Legendre transform (line) and from the exact eigenvalue histogram at $\varepsilon = 10^{-80}$ (dots); the von Neumann point is tangent to $f = \alpha$. Bottom right, for contrast: the quantum Hall wave-function spectrum $\Delta_q = 2q(1-q)[b_0 + b_1(q - \frac{1}{2})^2]$ of [8] and its parabolic approximation. This is a participation spectrum of $|\psi|^2$, not an entanglement spectrum.

Theorem 1 (Ideal Fermi-gas theorem). *Let ρ_A be Gaussian with entanglement energies $\{\varepsilon_k\}$, and set $n_q(\varepsilon) = (1 + e^{q\varepsilon})^{-1}$ and $s_F(n) = -n \ln n - (1 - n) \ln(1 - n)$. Then, exactly,*

$$\ln \text{Tr} \rho_A^q = \sum_k \left[\ln(1 + e^{q\varepsilon_k}) - q \ln(1 + e^{\varepsilon_k}) \right], \quad (4)$$

$$\tilde{S}_q = \sum_k s_F(n_q(\varepsilon_k)), \quad (5)$$

$$\langle K \rangle_q = \sum_k \left[\ln(1 + e^{\varepsilon_k}) - \varepsilon_k(1 - n_q(\varepsilon_k)) \right]. \quad (6)$$

In words, $\tilde{S}_q$ is the entropy of an ideal Fermi gas with single-particle levels ε_k at temperature $1/q$. If $\rho_A(\varepsilon) = \mathcal{L}^{-1} \sum_k \delta(\varepsilon - \varepsilon_k)$ has a limit, then $f_{\text{ent}}(\alpha_q) = \int d\varepsilon \rho_A(\varepsilon) s_F(n_q(\varepsilon))$.

Proof. ρ_A factorises over modes, with mode k having eigenvalues ζ_k and $1 - \zeta_k$, so $\text{Tr} \rho_A^q = \prod_k [\zeta_k^q + (1 - \zeta_k)^q]$. With $\zeta = (1 + e^\varepsilon)^{-1}$ one has $\zeta^q + (1 - \zeta)^q = (1 + e^{q\varepsilon})/(1 + e^\varepsilon)^q$, which gives (4). The escort state $\rho_A^q / \text{Tr} \rho_A^q \propto \exp(-q \sum_k \varepsilon_k c_k^\dagger c_k)$ is Gaussian with occupations $n_q(\varepsilon_k)$, and its

von Neumann entropy is $\sum_k s_F(n_q)$; by Proposition 2.1 this is $\tilde{S}_q$. Finally, $\langle K \rangle_q = -\partial_q \ln \text{Tr} \rho_A^q$, which gives (6). $\square$

Corollary 2.1. (i) $f_{\text{ent}} \geq 0$ and f_{ent} is concave, as is any microcanonical entropy of an ideal gas. (ii) f_{ent} depends on the state only through $\rho_A(\varepsilon)$. Each summand in (4) is exactly invariant under $\varepsilon \rightarrow -\varepsilon$ (particle-hole exchange of a mode), so f_{ent} depends only on the distribution of $|\varepsilon|$. (iii) A uniform density $\rho_A(\varepsilon) = 1/\pi^2$ per unit $\mathcal{L} = \ln \ell$ gives, using $\int_{\mathbb{R}} s_F((1+e^x)^{-1}) dx = \pi^2/3$,

$$f_{\text{ent}}(\alpha_q) = \frac{1}{3q}, \quad (7)$$

which is the Calabrese–Lefevre result $f(\alpha_q) = c/(3q)$ with $c = 1$ [4].

Remark (Bosonic Gaussian states). The same transcription holds for bosonic Gaussian states. By Williamson’s theorem the reduced state is $\rho_A = \bigotimes_k (1 - e^{-\varepsilon_k}) e^{-\varepsilon_k b_k^\dagger b_k}$, with symplectic eigenvalues $\nu_k = \frac{1}{2} \coth(\varepsilon_k/2)$ (vacuum normalised to $\nu = \frac{1}{2}$). Then

$$\ln \text{Tr} \rho_A^q = \sum_k \left[q \ln(1 - e^{-\varepsilon_k}) - \ln(1 - e^{-q\varepsilon_k}) \right], \quad \tilde{S}_q = \sum_k s_B(\bar{n}_q(\varepsilon_k)), \quad (8)$$

with $\bar{n}_q(\varepsilon) = (e^{q\varepsilon} - 1)^{-1}$ and $s_B(n) = (1+n) \ln(1+n) - n \ln n$. The refined Rényi entropy is thus the entropy of an ideal Bose gas at temperature $1/q$, by the same argument as in Theorem 1: the escort state is thermal with energies $q\varepsilon_k$. Pure modes ($\nu_k = \frac{1}{2}$) do not contribute, and $f_{\text{ent}} \geq 0$ is again concave. Unlike the fermionic case there is no $\varepsilon \rightarrow -\varepsilon$ symmetry, since $\varepsilon_k > 0$.

2.3 What wave-function multifractality can and cannot determine

The correlation matrix of a Slater determinant is $C = P_A \Pi P_A$, where $\Pi = \sum_{n \in \text{occ}} |\psi_n\rangle \langle \psi_n|$ and P_A projects onto A . Its non-trivial spectrum coincides with that of the overlap matrix $O_{nm} = \langle \psi_n | P_A | \psi_m \rangle$ restricted to occupied states.

Proposition 2.2 (Structure of the map from wave functions to entanglement). (i) f_{ent} is a functional of the spectrum of O , i.e. of joint statistics of overlaps between different eigenstates. Single-state moments $\langle |\psi_n|^{2q} \rangle$, and hence f_{wave} , do not determine it. For example, the particle-number variance $\kappa_2 = \text{Tr} O(1 - O) = \sum_{n \in \text{occ}, m \notin \text{occ}} |\langle \psi_n | P_A | \psi_m \rangle|^2$, which controls the leading entanglement [10], involves cross-energy correlations.

(ii) There is no rescaling under which f_{ent} coincides with a disorder-averaged f_{wave} on its full support. The latter can be negative and obeys $\Delta_q = \Delta_{1-q}$ [11], while $f_{\text{ent}} \geq 0$ (Corollary 2.1) and τ_{ent} has no $q \rightarrow 1 - q$ symmetry (for instance, $\tau \propto q - 1/q$ for a CFT).

Proof. (i) follows from $\text{spec}(P_A \Pi P_A) \setminus \{0\} = \text{spec}(\Pi P_A \Pi) \setminus \{0\}$ and Theorem 1. (ii) follows from the stated properties. $\square$

Numerical test. We diagonalised critical power-law random banded matrices (PRBM) [12] of the unitary class with $b = 1$ and $N = 2048$ (six realisations). We filled the lower half of the spectrum and took A to be half of the chain. The identity (5) holds to a relative accuracy of 5×10^{-10} . The wave-function spectrum of states near the band centre gives $D_2 \simeq 0.81$ and $\alpha_0 \simeq 1.11$ at this size. The entanglement spectrum, normalised by $S_1 = 13.7$, is non-negative everywhere, with minimum 0.026, and departs visibly from the clean Calabrese–Lefevre form (Fig. 2, top left). Disorder thus changes f_{ent} , but only through $\rho_A(\varepsilon)$, as Theorem 1 requires.

3 Markov random cascades in tensor networks

3.1 From random tensor networks to cascades

If a state can be written as $(V_A \otimes V_B) \bigotimes_{e \in \gamma} |\phi_e\rangle$ with isometries V_A, V_B and bond states $|\phi_e\rangle$ on a cut γ , then the non-zero spectrum of ρ_A is that of $\bigotimes_e \text{Tr}_e |\phi_e\rangle \langle \phi_e|$. For random tensor networks with large bond dimension, Rényi entropies are given by sums of bond Rényi entropies along the minimal cut [13]. For a MERA-like network [14], the cut of an interval crosses a bounded number of bonds per layer over $n \propto \log_b L$ layers. With identical bond spectra $\{p_j\}$, the spectrum of ρ_A is then a multinomial cascade, with

$$\tau(q) = -\frac{1}{\ln b} \ln \sum_j p_j^q \quad (\text{per layer, } \mathcal{L} = \ln L). \quad (9)$$

The continuum $f(\alpha)$ arises from the discrete layers through Cramér's theorem: $\ln \lambda$ is a sum of n i.i.d. terms, and f is the Legendre transform of their cumulant generating function (Appendix B). For a genuine MERA, disentanglers spoil the exact product structure, and the cascade is a model.

3.2 Markov modulation

To model residual correlations between layers, let the bond spectrum of layer k depend on a sector s_k that follows a Markov chain with transition matrix T and stationary distribution π . Define $z_q(s) = \sum_j p_{sj}^q$ and $D_q = \text{diag}(z_q)$. The annealed partition function over n layers is $Z_n(q) = \pi^\top D_q (TD_q)^{n-1} \mathbf{1}$.

Proposition 3.1 (Perron-root mass exponent). *For irreducible aperiodic T , $\tau(q) = -\ln \Lambda_q / \ln b$, where Λ_q is the Perron root of TD_q . Λ_q is real-analytic in q , so τ and f have no singularities at finite q .*

Proof. TD_q is non-negative, irreducible and aperiodic, so its Perron root is simple and strictly dominant, and $\frac{1}{n} \ln Z_n \rightarrow \ln \Lambda_q$. The matrix entries are analytic in q , and a simple isolated eigenvalue of an analytic matrix family is analytic [15]. $\square$

Proposition 3.2 (Two-state chain). *For the symmetric two-state chain with second eigenvalue λ_2 and $\pi = (\frac{1}{2}, \frac{1}{2})$, writing $S = z_1 + z_2$ and $P = z_1 z_2$,*

$$\Lambda_q = \frac{1}{4} \left[(1 + \lambda_2) S + \sqrt{(1 + \lambda_2)^2 S^2 - 16 \lambda_2 P} \right], \quad (10)$$

$$\Delta\tau(q) \equiv \tau(q) - \tau(q)|_{\lambda_2=0} = -\frac{\lambda_2}{\ln b} \frac{\text{Var}_\pi(z_q)}{\langle z_q \rangle_\pi^2} + O(\lambda_2^2). \quad (11)$$

Proof. With $T = \begin{pmatrix} 1-e & e \\ e & 1-e \end{pmatrix}$ and $\lambda_2 = 1 - 2e$, the matrix TD_q has trace $\frac{1+\lambda_2}{2} S$ and determinant $\lambda_2 P$, which gives the first formula. Differentiating at $\lambda_2 = 0$ gives $\partial_{\lambda_2} \Lambda_q = (z_1 - z_2)^2 / (2S) = \text{Var}_\pi(z_q) / \langle z_q \rangle_\pi$. $\square$

Two features distinguish (11) from a naive guess $\Delta\tau \propto \lambda_2^q$. The correction is *linear* in λ_2 , because λ_2 weights a single transition per layer irrespective of q , and q enters only through $\text{Var}_\pi(z_q)$. Moreover $z_1(1) = z_2(1) = 1$, so $\Delta\tau(1) = 0$ and normalisation is preserved.

Phase transitions. By Proposition 3.1 there are none for $\lambda_2 < 1$. At $\lambda_2 = 1$ the chain is reducible and $\Lambda_q = \max(z_1, z_2)$. This has a kink wherever $z_1(q) = z_2(q)$ with different slopes, and in particular at $q = 1$ whenever the two sectors have different von Neumann entropies. The Legendre transform then contains a linear segment, i.e. a Maxwell construction and a first-order transition. For $\lambda_2 \rightarrow 1$ the discriminant shows that the crossover sharpens on the scale $|z_1 - z_2| \sim (1 - \lambda_2)^{1/2} S$. Because the partition function is annealed over sectors, f may become negative, by the same mechanism that makes disorder-averaged wave-function spectra negative. A single ρ_A always has $f \geq 0$.

Numerical test. With bond spectra $(0.7, 0.3)$ and $(0.9, 0.1)$ and $b = 2$, the first-order formula reproduces the exact Perron result with relative error 2.1×10^{-2} at $\lambda_2 = 0.05$ and 8.4×10^{-2} at $\lambda_2 = 0.2$, i.e. with error $O(\lambda_2^2)$. A finite transfer-matrix product with $n = 400$ layers gives 0.6197 against the Perron value 0.6194 at $q = 2.5$, $\lambda_2 = 0.5$. The sharpening of the crossover as $\lambda_2 \rightarrow 1$ is shown in Fig. 2 (top right).

4 Non-parabolicity and Casimir invariants at the quantum Hall critical point

4.1 Weyl invariance and the Casimir variable

At Anderson transitions the multifractal exponents obey $\Delta_q = \Delta_{1-q}$ [11]. Gruzberg, Ludwig, Mirlin and Zirnbauer traced this symmetry to the invariance of the nonlinear σ -model under the Weyl group of its target space [16]. Scaling operators are labelled by highest weights, and at one loop their dimensions are proportional to the quadratic Casimir eigenvalue [17]. For the operators $|\psi|^{2q}$ the Weyl reflection acts as $q \rightarrow 1 - q$. Any Weyl-invariant function of q is therefore a function of

$$u = q(1 - q). \quad (12)$$

Proposition 4.1 (Casimir form of the quantum Hall spectrum). *The high-precision quantum Hall result [8], $\Delta_q = 2q(1 - q)[b_0 + b_1(q - \frac{1}{2})^2]$ with $b_0 = 0.1291 \pm 0.0002$ and $b_1 = 0.0029 \pm 0.0003$, is identical to*

$$\Delta_q = c_1 u + c_2 u^2, \quad c_1 = 2(b_0 + \frac{b_1}{4}) = 0.2596, \quad c_2 = -2b_1 = -0.0058. \quad (13)$$

In particular $\alpha_0 - 2 = \Delta'_0 = c_1$, in agreement with the reported $\alpha_0 = 2.2596$.

Proof. Use $(q - \frac{1}{2})^2 = \frac{1}{4} - u$, which gives $2u[b_0 + b_1(\frac{1}{4} - u)] = 2(b_0 + \frac{b_1}{4})u - 2b_1 u^2$. Also $\Delta'_0 = c_1$, since $u'(0) = 1$ and $u(0) = 0$. $\square$

A parabolic spectrum is thus exactly the statement that Δ_q is linear in the quadratic Casimir. The non-parabolicity b_1 measures the u^2 term, i.e. the contribution of higher Casimir invariants.

4.2 Origin in the loop expansion

In $2 + \varepsilon$ dimensions the unitary-class result through four loops is [3]

$$\Delta_q = q(1 - q) \left(\frac{\varepsilon}{2} \right)^{1/2} - \frac{3}{8} \zeta(3) q^2 (q - 1)^2 \varepsilon^2 + O(\varepsilon^{5/2}) = u \left(\frac{\varepsilon}{2} \right)^{1/2} - \frac{3}{8} \zeta(3) u^2 \varepsilon^2 + \dots \quad (14)$$

The u^2 term first appears at four loops, through the $\zeta(3)$ contribution. It gives $c_2 = -\frac{3}{8} \zeta(3) \varepsilon^2 < 0$, that is,

$$b_1 = \frac{3}{16} \zeta(3) \varepsilon^2 > 0, \quad (15)$$

which has the same sign as the measured value. We stress that the $2 + \varepsilon$ expansion is not controlled at the IQH fixed point, which lies at strong coupling with topological angle $\theta = \pi$. The loop expansion therefore explains the *type* and *sign* of the non-parabolic term, but not its magnitude.

4.3 Incompatibility with current-algebra primaries

Proposition 4.2. *If the operators $|\psi|^{2q}$ at the IQH fixed point were primaries of an affine current algebra realised through the Sugawara construction (as in a WZW model), their dimensions would be proportional to the quadratic Casimir, and b_1 would vanish.*

Proof. For current-algebra primaries in a representation R , the Sugawara construction gives $h_R = C_2(R)/(k + h^\vee)$ [18]. The dimension is then a linear function of C_2 , hence of u , and $c_2 = 0$. $\square$

Since $b_1 = 0.0029 \pm 0.0003$ differs from zero by about ten standard deviations [8], theories in which the multifractal operators are Sugawara primaries are excluded. This does not exclude a WZW model perturbed by an operator that breaks the current algebra, nor a description in which $|\psi|^{2q}$ are not current-algebra primaries. We do not know which operator generates the u^2 term at the IQH fixed point.

5 Holographic duals and higher-curvature constraints

5.1 Cosmic branes and the two-dimensional case

In holographic CFTs the refined Rényi entropy is the area of a cosmic brane of tension $T_q = (q - 1)/(4qG_N)$ [6]. For spherical regions the brane is the horizon of a hyperbolic black hole at temperature T_0/q [19]. In $d = 2$ the q -dependence of Rényi entropies is universal for every CFT, $S_q = \frac{c}{6}(1 + \frac{1}{q})\ln(\ell/\varepsilon)$. The corresponding spectrum is $f(\alpha) = \sqrt{\frac{c}{3}(2\alpha - \frac{c}{3})}$ [4], which is of square-root type rather than parabolic and has no maximum: $\alpha \rightarrow \infty$ as $q \rightarrow 0$. In a three-dimensional bulk the Gauss–Bonnet term vanishes identically, so higher-curvature terms of this type cannot change the two-dimensional result.

5.2 Gauss–Bonnet branes in $d = 4$

For Gauss–Bonnet gravity in $d + 1 = 5$ dimensions, define $a = \lambda_{\text{GB}} f_\infty = \frac{1}{2}(1 - \sqrt{1 - 4\lambda_{\text{GB}}})$. The temperature and horizon entropy of the hyperbolic black hole with horizon radius x , in units of the AdS radius [20], are

$$\frac{T(x)}{T_0} = \frac{d(1 - a)x^4 - (d - 2)x^2 + (d - 4)a}{2x(x^2 - 2a)}, \quad \frac{S(x)}{S_1} = x^{d-1} \frac{1 - \frac{2a(d-1)}{(d-3)x^2}}{1 - \frac{2a(d-1)}{d-3}}. \quad (16)$$

Here $S(x)$ is the Jacobson–Myers horizon entropy, which replaces the area in Dong’s prescription for this theory [21]. The brane sits at x_q , defined by $T(x_q)/T_0 = 1/q$. Define $I(q) = \int_{x_q}^1 \frac{S}{S_1} d(\frac{T}{T_0})$. The thermodynamic method of [19] then gives, exactly,

$$\tau(q) = q I(q), \quad \alpha_q = I(q) + \frac{S(x_q)}{q S_1}, \quad f(\alpha_q) = \frac{S(x_q)}{S_1}, \quad (17)$$

in units of S_1 . The last relation is Dong’s formula, which is automatic in this construction.

C_T universality. Writing $\tau(q) = (q - 1) + t_2(q - 1)^2 + \dots$, Perlmutter’s theorem [22] fixes t_2 in terms of C_T relative to the normalisation of S_1 . For Gauss–Bonnet gravity in $d = 4$ one has $C_T \propto 1 - 2a$ and $a_{\text{anom}} \propto 1 - 6a$ [23], so $t_2^{\text{GB}}/t_2^{\text{E}} = (1 - 2a)/(1 - 6a)$. Table 1 confirms this numerically to about 10^{-3} . In the Einstein limit, the numerically integrated S_q agrees with the closed form of [19] to 3×10^{-10} .

λ_{GB}	a	t_2	t_2/t_2^{E}	$(1-2a)/(1-6a)$
$-7/36$	-0.1667	-0.3303	0.6672	0.6667
-0.10	-0.0916	-0.3782	0.7639	0.7635
0	0	-0.4951	1	1
0.05	0.0528	-0.6479	1.3085	1.3090
0.09	0.1000	-0.9895	1.9985	2.0000

Table 1: Second Taylor coefficient of the holographic mass exponent for Gauss–Bonnet gravity in $d = 4$, over the allowed range $-7/36 \leq \lambda_{\text{GB}} \leq 9/100$.

Negative refined Rényi entropy at finite q . For $a > 0$ the Jacobson–Myers entropy changes sign at $x^2 = 2a(d-1)/(d-3)$. As $q \rightarrow \infty$, $x_q^2 \rightarrow (d-2)/(d(1-a))$ for $d = 4$. At $\lambda_{\text{GB}} = 0.09$ we find that $f(\alpha_q) = \tilde{S}_q/S_1$ becomes negative for $q > q_c \simeq 6.47$ (Fig. 2, bottom right). Because $\tilde{S}_q$ is the entropy of a density matrix, it cannot be negative. The hyperbolic black-hole saddle therefore cannot dominate beyond q_c , and another saddle, or a breakdown of the effective description, must take over. This is a saddle-exchange transition of the kind discussed in Section 3.

5.3 No dictionary between b_1 and λ_{GB}

It is tempting to relate the quantum Hall coefficient b_1 to a holographic coupling such as λ_{GB} . We argue that this would be unfounded, for three reasons.

1. The measures differ: a disorder-averaged participation spectrum (f may be negative, $\Delta_q = \Delta_{1-q}$) versus the entanglement spectrum of a clean, strongly coupled CFT ($f \geq 0$, no $q \rightarrow 1-q$ symmetry).
2. The dimensions differ: the IQH transition is two-dimensional, where holographic Rényi entropies are universal, whereas Gauss–Bonnet effects require $d \geq 4$.
3. The Weyl-invariant variable u that organises the quantum Hall spectrum has no counterpart in τ_{ent} .

What the two problems share is an organising principle. In each case the departure from the leading form is controlled by the next invariant of the critical theory: the u^2 (higher-Casimir) term for localisation, and CFT data such as C_T/a and $\langle TTT \rangle$ for holography.

6 Discussion and limitations

1. Theorem 1 and the subsequent Remark are exact for fermionic and bosonic Gaussian states, respectively. For interacting systems the modular Hamiltonian is not quadratic, and f_{ent} is no longer an ideal-gas entropy.
2. Proposition 2.2 is a structural statement. We have not derived $\rho_A(\varepsilon)$ from the multi-state correlations of critical wave functions; doing so would require the joint statistics of overlaps across energies, beyond the single-state exponents Δ_q .
3. The PRBM numerics use a single system size, so the extracted exponents ($D_2 \simeq 0.81$, $\alpha_0 \simeq 1.11$) carry finite-size corrections that we have not quantified.
4. The Markov cascade is a phenomenological model of residual inter-layer correlations in tensor networks, and its partition function is annealed. Quenched (typical) averages may exhibit freezing-type transitions that we have not analysed.

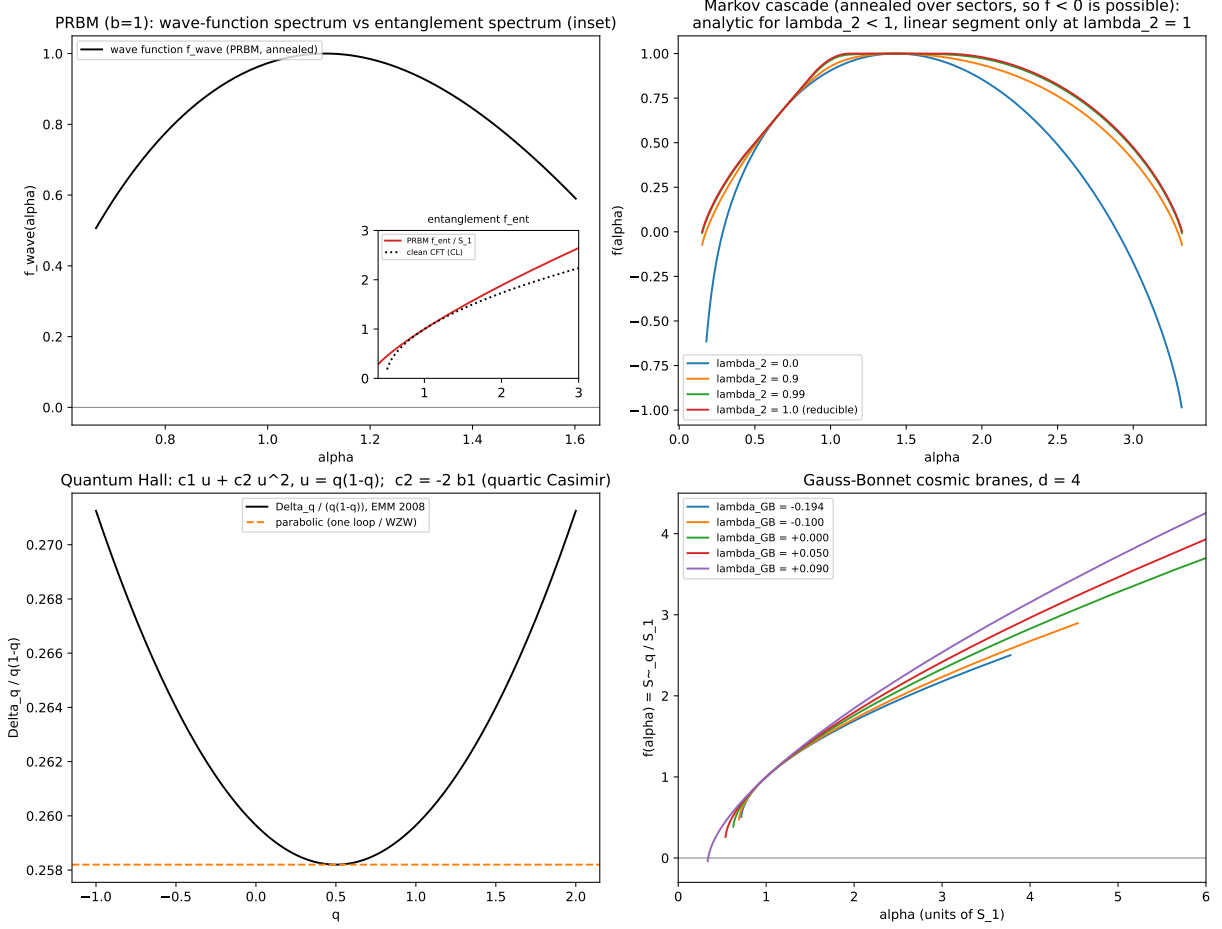


Figure 2: Top left: wave-function singularity spectrum of a critical PRBM ($b = 1$, $N = 2048$, annealed over states near the band centre); inset: entanglement spectrum f_{ent}/S_1 of the half-filled Slater determinant compared with the clean Calabrese–Lefevre form. Top right: Markov-modulated cascade, annealed over sectors, for $\lambda_2 = 0, 0.9, 0.99, 1$; the spectrum is analytic for $\lambda_2 < 1$ and develops a linear segment at $\lambda_2 = 1$. Bottom left: quantum Hall $\Delta_q/q(1-q)$ [8] against the parabolic value; the curvature is the u^2 coefficient $c_2 = -2b_1$. Bottom right: Gauss–Bonnet cosmic-brane spectra in $d = 4$.

5. Section 4 identifies the type and sign of the non-parabolic term, but the $2 + \varepsilon$ expansion is not controlled at the IQH fixed point. The operator generating the u^2 term remains unknown.
6. The negative refined Rényi entropy for $\lambda_{\text{GB}} > 0$ indicates a change of dominant saddle; we have not constructed the competing saddle.

Data and code availability

All figures and numbers are produced by two Python scripts, `multifractal_entanglement.py` (Section 2, Fig. 1) and `frontier2_deepdive.py` (Sections 2–5, Fig. 2, Table 1), available at <https://github.com/Ruqing1963/multifractal-quantum-entanglement> and archived on Zenodo under DOI 10.5281/zenodo.23193512.

A Numerical methods

- *Two-scale Cantor measure.* $\tau(q)$ is obtained from $\sum_i p_i^q r_i^{-\tau} = 1$ by Brent’s method, with $\alpha = \sum_i w_i \ln p_i / \sum_i w_i \ln r_i$ and $w_i = p_i^q r_i^{-\tau}$. The exact spectrum at resolution ε is enumerated by composition (k_1, k_2) with the stopping rule “first prefix of size $\leq \varepsilon$ ”, using logarithmic binomial coefficients. The trace equals 1 to 10^{-12} . At $\varepsilon = 10^{-80}$, $S_q/\mathcal{L}$ differs from D_q by $O(1/\mathcal{L})$, e.g. 0.5633 against 0.5616 at $q = 1$.
- *PRBM.* H_{ij} is complex Gaussian with variance $[1 + (r_{ij}/b)^2]^{-1}$, using the periodic distance r_{ij} , and is Hermitised. C is formed from the lower half of the eigenvectors. Wave-function moments use box sizes $2, 4, \dots, N/8$ and the 64 states closest to the band centre. Entanglement quantities use $\ln(\zeta^q + (1 - \zeta)^q)$ with ζ clipped to $[10^{-15}, 1 - 10^{-15}]$.
- *Markov cascade.* The Perron root is computed by dense diagonalisation of TD_q . The finite- n check multiplies normalised row vectors for $n = 400$ layers.
- *Gauss–Bonnet.* x_q is found by Brent’s method above the largest zero of the numerator of $T(x)$. $I(q)$ is computed by adaptive quadrature, and α_q from the exact relation in Section 5. t_2 is obtained by central differences on a uniform grid in q .

B Large-deviation derivations

Equal-scale cascade (Cramér). With n independent layers and bond spectrum $\{p_j\}$, the eigenvalues are $\lambda = \prod_k p_{j_k}$. Their number with $-\ln \lambda / (n \ln b) \in [\alpha, \alpha + d\alpha]$ is a multinomial count. By Stirling’s formula, or equivalently Cramér’s theorem for the empirical frequencies [24], it behaves as $b^{nf(\alpha)}$, with f the Legendre transform of $\tau(q) = -\log_b \sum_j p_j^q$.

Markov modulation (Gärtner–Ellis). For an irreducible Markov additive process the scaled cumulant generating function exists and equals $\ln \Lambda_q$, the logarithm of the Perron root of the tilted matrix [24]. Since it is differentiable, the Gärtner–Ellis theorem yields the large-deviation rate function as its Legendre transform, which is f up to normalisation.

Crossover width near $\lambda_2 = 1$. For $\lambda_2 = 1 - 2e$ the discriminant in Proposition 3.2 is $4(z_1 - z_2)^2 + O(e)S^2$. The smooth Λ_q therefore deviates from $\max(z_1, z_2)$ appreciably only where $(z_1 - z_2)^2 \lesssim eS^2$, i.e. on a window of width $\propto (1 - \lambda_2)^{1/2}$ around each crossing.

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